Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/02/2025
PI Name:	Morris, Sarah Ph.D.		
Project Title:	Analysis of Passive Phase Separations and Bubble Dynamics in Developing Shear Flows in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	59717	**Congressional District:**	1
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2023 NNH23ZDA001N-PSI E.8 Physical Sciences Informatics
Start Date:	06/01/2025	**End Date:**	05/31/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Period of performance changed to 06/01/2025 - 05/31/2027 per NSSC documentation. Original period of performance was 10/01/2024 - 09/30/2026 (Ed., 10/27/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	The Capillary Channel Flow Phase Separation Experimental Unit 2 (CCF-PS-EU2) was launched to the International Space Station (ISS) in 2010. These experiments study capillary-induced bubble migrations in developing shear flows, and demonstrate passive phase separations in microgravity. This data is available in the NASA Physical Sciences Informatics (PSI) database as a hyperlinked file set of video clips. Previous data reduction efforts on the CCF-PS-EU2 dataset rely almost exclusively on analyses of these videos and have been employed to construct dozens of regime maps and verify numerical and theoretical analyses. However, despite the enormity of the data archive (≈ 25,000 data sets), the published works have not addressed several key aspects of the flow that are of both fundamental scientific and applied engineering value, including the net effects of subtle forces of fluid dynamics origins for these unearthly highly-deformable migrating bubbles. This is in large part due to inexhaustive experimental analyses that do not make use of the most modern particle tracking techniques and specific challenges for the applied numerical modeling. In this work, we propose a numerical and experimental analysis of the CCF-PS-EU2 dataset to compare to existing and developing theory. We propose to improve all aspects of previous efforts; firstly, we will perform full 3D simulations of the complex flow structures in this system using proven, open-source computational frameworks that we have developed over the past 4 years. We will also analyze the video dataset and quantify spatial-temporal bubble motions with improved resolution and precision via state-of-the-art particle tracking velocimetry (PTV) techniques. It is planned to develop new predictions of bubble-inscribed locations that take into account both geometry (existing theory) and physical flow phenomena (theory to be developed). The proposed computational and video analysis methods have been demonstrated in an adjacent PSI database (the Interior Corner Flow PSI experiments); and, as such, the research team is familiar with both the PSI data structure and methods to extract quantitative results from raw video data. Understanding passive phase separation dynamics and their response to channel geometries offers design advantages in fuel control, cryogenic liquid management, medical fluids, and life-support systems aboard orbiting spacecraft. The research team's experience – with the PSI database system, quantitative video processing and analysis, and the computational framework – will yield immediate results ready for dissemination via publications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2025
Bibliography Type:	Description: (Last Updated:)